

 Did You Know?

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data without being explicitly programmed.

 What is Machine Learning?

Machine Learning (ML) is a field of Artificial Intelligence that allows computers to learn from experience and improve their performance automatically using data.

 Why do we use Machine Learning?

- Analyze large amounts of data
- Predict future outcomes
- Improve decision making
- Automate repetitive tasks
- Increase accuracy

Types of Machine Learning

Type	Description
Supervised Learning	Uses labeled training data.
Unsupervised Learning	Uses unlabeled data to find patterns.
Reinforcement Learning	Learns through rewards and penalties.

Supervised Learning

Supervised Learning trains a model using labeled data where both input and correct output are known.

Input → Model → Correct Output



Unsupervised Learning

Unsupervised Learning works with unlabeled data. It discovers hidden patterns and groups automatically.

Customer Data



Grouping



Customer Clusters



Reinforcement Learning

Reinforcement Learning allows an agent to learn by interacting with the environment and receiving rewards or penalties.

Action



Reward



Better Decision



Applications of Machine Learning

- Recommendation Systems
- Email Spam Detection
- Face Recognition

- Medical Diagnosis
- Fraud Detection
- Self-driving Cars

Advantages

- Improves with experience
- Handles large datasets
- High prediction accuracy
- Automates decision making
- Saves time

Disadvantages

- Requires large datasets
- Training can be slow

- High computational cost
- May produce biased results
- Difficult to interpret complex models



Interview Questions

Q1: What is Machine Learning?

Machine Learning is a branch of AI that enables computers to learn from data.

Q2: Name the three types of Machine Learning.

Supervised, Unsupervised and Reinforcement Learning.

Q3: Which learning method uses labeled data?

Supervised Learning.

Q4: Which learning method uses rewards and penalties?

Reinforcement Learning.

Quick Revision

ML Learning from Data

Supervised Labeled Data

Unsupervised Unlabeled Data

Reinforcement Reward & Penalty